Roadmap. §45 (pp. 1–3): the arrow on fermion lines, from LecNote17 p. 15's $Z_0 = \exp[i\int\bar\eta S\eta]$, and the Yukawa trees $e^\mp\varphi$, $e^+e^-\to\varphi\varphi$, $e^\mp e^\mp$. §§46–48 (pp. 4–6): the unpolarized $|\mathcal{T}|^2$ as a trace; "page 10–12" is LecNote17 pp. 10–12 (§47). Next, LecNote21: the photon field (§§54–57); this trace route returns in LecNote23 ($e^+e^-\to\gamma\gamma$). Supplements: DiracFeynmanRules, FermionSigns, DiracSpinors, GammaTraces, CrossSection_WorkedExample (flux, dPS_2).

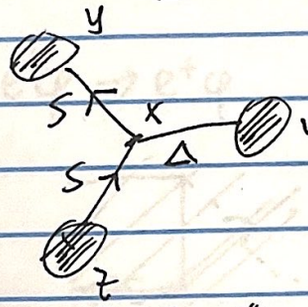
lec 20

But there is additional rules:
note that "charge is conserved" such that
each propagator is $\bar\eta(x) S(x-y)\eta(y)$

$$\frac{\delta}{\delta\bar\eta(x)}\frac{\delta}{\delta\eta(y)} \;\&\; \bar\psi(x)\psi(y)$$

each vertex is $-ig\int d^4x\ \bar\psi(x)\psi(x)\varphi(x)$

$\eta(x)$ $\bar\eta(x)$



$$-ig\int d^4x\,d^4y\,d^4z\,d^4w\ \bar\eta(y)\,S(y-x)\,S(x-z)\,\Delta(z-w)\,J(w)$$ [1]

$\bar\eta(x),\eta(x)$ Removed by functional derivative

1 Should read $\Delta(x-w)$: φ attaches at the vertex x, as drawn; the vertex's $\frac{1}{i}\frac{\delta}{\delta J(x)}$ on $e^{\frac{i}{2}\int J\Delta J}$ gives $\int\Delta(x-w)J(w)$. z is the η source point, not a vertex.

So there is a "direction" which is from $\bar\eta\to\eta$, [2]
(left to right) in the Equation $\bar\psi\to\eta$

and we can set conveniently choose to be with the flow of charge.

2 The arrow, as drawn ($z\to x\to y$), runs from the η end to the $\bar\eta$ end: an e^- enters at a $\bar\Psi$ ($b^\dagger_{\text{in}}$) and leaves at a Ψ (b_{out}), so $Q = N_b - N_d$ flows along it, and the formula is read from the head back, $\bar u_{\text{out}}\cdots u_{\text{in}}$. Charge conservation forbids $\langle\Psi\Psi\rangle$: one arrow in, one out per vertex.

(e^-)

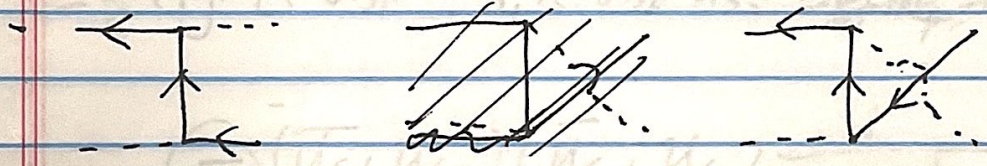
The ψ-type momentum "flow" is the same as charge and $\bar{\psi}$-type opposite, (just a notational choice).

(e^+)

Now we have

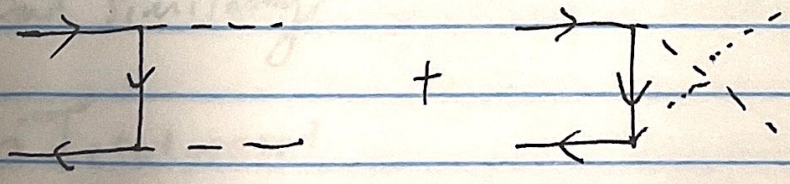
$$i\mathcal{T}_{e^-\varphi\to e^-\varphi} = \frac{1}{i}(ig)^2\,\bar{u}_{s_1}(\vec{p}')\left[\frac{-\not{p}-\not{k}+m}{(p+k)^2+m^2-i\varepsilon} + \frac{-\not{p}+\not{k}'+m}{(p-k')^2+m^2-i\varepsilon}\right]u_s(\vec{p})$$

Let's check $e^+\varphi \to e^+\varphi$.



$$i\mathcal{T}_{e^+\varphi\to e^+\varphi} = \frac{1}{i}(ig)^2\,\bar{v}_s(\vec{p})\left[\frac{-\not{p}-\not{k}+m}{(\not{p}+\not{k})+m^2-i\varepsilon} + \frac{-\not{p}+\not{k}'+m}{(\not{p}-\not{k}')^2+m^2-i\varepsilon}\right] v_{s_1}(\vec{p}')$$ (1)

Now $e^+e^-\to\varphi\varphi$



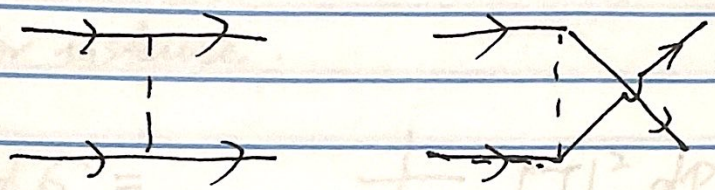
$$i\mathcal{T}_{e^+e^-\to\varphi\varphi} \quad (p_1,p_2)\ (k_1,k_2) \qquad = \frac{1}{i}(ig)^2\,\bar{v}_{s_2}(\vec{p}_2)\left(\frac{-\not{p}_1-\not{k}_1+m}{-s+m^2-i\varepsilon} + \frac{-\not{p}_1-\not{k}_2+m}{-u+m^2-i\varepsilon}\right)u_{s_1}(\vec{p}_1)$$ (2)

1 Numerators should be $\not{p}+\not{k}+m$ and $\not{p}-\not{k}'+m$: the incoming e^+ sits at a Ψ point, so it moves against the arrow (not a choice), which carries $-(p+k)$, $-(p-k')$; $\tilde{S}(-q) = (\not{q}+m)/(q^2+m^2)$. Check: same spin-summed $|\mathcal{T}|^2$ as $e^-\varphi$ (C).

2 Should read $\bar{v}_2[\tilde{S}(p_1-k_1)+\tilde{S}(p_1-k_2)]u_1$: numerators $-\not{p}_1+\not{k}_{1,2}+m$, denominators $-t+m^2$, $-u+m^2$ with $t=-(p_1-k_1)^2$ (the t, u graphs drawn; e^- has p_1). Then $k_1\leftrightarrow k_2$ symmetric, as identical scalars require.

And finally 4 fermions

$e^- e^- \to e^- e^-$



$$i\mathcal{T}_{e^-e^-\to e^-e^-} = \frac{1}{2}(ig)^2 \bar{u}_{s_1'}(p_1') \cdots u_{s_1}(p_1)$$

$$= \frac{1}{2}(ig)^2 \Big[(\bar{u}_{s_1'} u_{s_1})(\bar{u}_{s_2'} u_{s_2}) \frac{1}{t - m_\varphi^2 + i\varepsilon} - (\bar{u}_{s_2'} u_{s_1})(\bar{u}_{s_1'} u_{s_2}) \frac{1}{u - m_\varphi^2 + i\varepsilon} \Big]$$

↓

exchange of fermions in the (final) state.[1]

And similarly

$$i\mathcal{T}_{e^+e^+\to e^+e^+}$$

$$= \cdots (\bar{v}_{s_1} v_{s_1'})(\bar{v}_{s_2} v_{s_2'}) \frac{1}{-t + m_\varphi^2 - i\varepsilon}$$

$$- (\bar{v}_{s_1} v_{s_2'})(\bar{v}_{s_2} v_{s_1'}) \frac{1}{-u + m_\varphi^2 - i\varepsilon}$$

1 Only relative signs are physical: reordering the $b^\dagger$'s in a state flips every term. Read each chain from its arrow head; terms whose spinor orders $(1'1\,2'2,\ 2'1\,1'2)$ differ by an odd permutation get -1. So $e^- e^+ \to e^- e^+$ is t-channel minus s-channel annihilation, and a closed fermion loop gives -1 and a trace (DiracFeynmanRules, FermionSigns).

~~For~~ In principle, we can calculate for observables now.

For instance.

$$d\sigma_{e^-_{s_1}\varphi \to e^-_{s_3}\varphi} = \frac{1}{\text{flux}} |\mathcal{T}|^2 \, dPS_2$$

$$\mathcal{T}_{ss'} = g^2 \bar{u}_{s'}(\vec{p}') \left[\frac{-\not{p} - \not{k} + m}{-s + m^2} + \frac{-\not{p} + \not{k}' + m}{-u + m^2} \right] u_s(\vec{p}).$$

plug in the expressions

plug in the expressions

You will get the matrix element for ~~one to use~~ a particular spin choices.

It is more often more convenient to ~~write~~ choose the spin quantization axis along the momentum direction, which is called helicity.

You will see in your HW it coincides with the chiral states in massless limit.

(if you work slightly more!)

1 Each $\mathcal{T}_{ss'}$ depends on the spin axis (helicity is one choice) and needs the explicit u's (DiracSpinors). A rate that ignores spin, $\frac{1}{2}\sum_{s,s'} |\mathcal{T}_{ss'}|^2$, does not: $\sum_s u_s \bar{u}_s = -\not{p} + m$ is the same for any axis (LecNote18 p. 7's boxed remark). Pages 5–6 use exactly this: no spinors, only a trace.

But if we, experimentally, prepare only the unpolarized initial state and measure the rate in total (e.g., detector does not measure spin, which is typically in high energy), many things are further simplified.

Let's see.

$$\overline{u}_s \beta A^* \beta u_{s'} = \overline{u}_s \bar{A} u_{s'}$$

$$|\mathcal{M}|^2 = g^4 \, \overline{u}_{s'} A u_s \, \overline{u}_s \bar{A} u_{s'} \, \boxed{u_s^* A^{\dagger *} \beta^* u_{s'}}$$

$\bar{A} = A$ for this particular one. [1]

$$A = \frac{-\not{p} + m}{\#}$$

$$A^* = \frac{-\not{p}^* + m}{\#}$$

$$\bar{A} = \beta A^* \beta = \frac{-\beta \not{p}^* \beta + m\beta^2}{\#}$$

$\beta^2 = \mathbb{1}$.

$$\not{p}^* \gamma^{\mu *} = \gamma^0 \gamma^\mu \gamma^0$$

$$= g^4 \, \overline{u}_{s'} A u_s \overline{u}_s A u_{s'}$$

$$= g^4 \, \mathrm{Tr}\left[u_{s'} \overline{u}_{s'} A u_s \overline{u}_s A \right]$$

[1] Here $*$ means $\dagger$ (the box and the last line are overwritten so). As $\bar{u}\beta = u^\dagger$, $(\bar{u}' A u)^* = u^\dagger A^\dagger \beta u' = \bar{u} \bar{A} u'$, $\bar{A} \equiv \beta A^\dagger \beta$. Since $\beta \gamma^{\mu\dagger} \beta = \gamma^\mu$, $\bar{A} = A$ for real combinations of 1 and γ^μ ($i\epsilon$ dropped). Literal complex conjugation fails: $\beta \gamma^{1*} \beta = -\gamma^1$.

Note that $\sum_{s=\pm} u_s(\vec{p})\,\bar{u}_s(\vec{p}) = -\not{p} + m$

$\sum_{s=\pm} v_s(\vec{p})\,\bar{v}_s(\vec{p}) = -\not{p} - m$.

$\sum_{s,s'} |T|^2 = \frac{1}{4} \sum_{s,s'} \mathrm{Tr}[(-\not{p}_1 - m)\,A\,(-\not{p} + m)\,A]$ (1)

Now. γ – gymnastics.

(go back to page 10-12)

$= \mathrm{Tr}[(\not{p}_1 + m)\left(\frac{(-\not{p} - \not{k} + m)}{-s + m^2} \cdot + \frac{(-\not{p} - \not{k}_1 + m)}{-u + m^2}\right)$ (2)

$(\not{p} - m)($ $)]$.

Only two types

$\mathrm{Tr}[\not{p}\not{k}]\,m^2 + \mathrm{Tr}[\not{p}'\not{p}\not{p}\not{k}]$ (3)

$\not{p}\not{k}_1$ ↓ p^2.

1 Should read $g^4\,\mathrm{Tr}[(-\not{p}' + m)A(-\not{p} + m)A]$: $\sum_{s'} u_{s'}\bar{u}_{s'} = -\not{p}' + m$, line 1. So the next line starts $(\not{p}' - m)$. Line 2: $\sum_s v_s\bar{v}_s = -\not{p} - m$ (the overwritten sign is $-$).

2 Should read $-\not{p} + \not{k}' + m$, as on p. 4: the u-channel line carries $p - k'$. This slip and the first-factor slip above each change the result.

3 Should read $\not{p}\,\not{p} = -p^2 = m^2$. For an unpolarized e^-, $\langle|\mathcal{T}|^2\rangle = \frac{1}{2}\sum_{s,s'}|\mathcal{T}|^2$; GammaTraces finishes the trace: $\frac{113}{64}g^4$ at $m = M = 1$, $s = 9$, $u = -1$.